

Wireless Boom

Global warming is a hot topic in today's news. People are seeking ways to save energy and protect the environment. Reducing household energy consumption is one effective way to slow the trend toward global warming. Utilities have successfully implemented demand response programs for commercial and industrial consumers. Now more and more energy companies, such as San Diego Gas & Electric (SDG&E) and Burbank Power and Water (BPW), are switching residential consumers' old mechanical meters to new smart digital meters, in order to implement demand response programs in the residential sector (see "Developing Affordable Smart Thermostats," p. 16). These smart meters connect the utility to the home over a network. They can help consumers monitor and control their energy consumption, and help utilities decrease peak demand, reducing the need to use or build more inefficient and dirty power plants that come online only to meet peak loads.

Several mature wireless network technologies are available to communicate between ambient digital devices in homes and smart meters. Some of the potential applications of smart meters and wireless networking products are described below.

Smart appliances (and smarter consumers). Appliances connected to a wireless network could provide a number of advanced connections between appliances and the utility company, creating enhanced services to monitor energy consumption and cost. For instance, using wireless technology, a network would notice how much energy is being consumed by an air conditioner during peak hours in the summer. This information could be supplied to an in-home display or a Web site, allowing consumers or utilities to adjust the air conditioner's energy consumption

automatically. Based on information provided by smart appliances, utilities could, in turn, tailor programs to meet the needs of consumers whose homes use a combination of critical and noncritical peak pricing load appliances.

Public safety services. Wireless-networked homes have the added benefit of providing safety and protection for occupants and their assets through remote monitoring devices, such as smoke detectors that trigger safety scenarios. When the wireless smoke detectors detect fire, the HVAC system immediately turns off to prevent smoke from circulating, and to prevent a flow of fresh oxygen from feeding the fire. Lights come up to a dimmed level—not too bright to blind the people they awaken—to help occupants find their way through the dark and out of the house. Motorized blinds and the garage door open automatically to provide quick escape. The home intercom system even tells the occupants the best escape route. The exterior lights also flash to make the house easier for the firefighters to locate.

Home automation. Wireless networks can detect water or gas leaks, freezing pipes, and fire; and can contact the owner, the fire department, or a plumber.

SDG&E is actively exploring using ZigBee wireless technology (sends signals over short distances, like Bluetooth) and its home area network (HAN) program, and the utility recently partnered with the University of California, San Diego, to complete a successful pilot project using this technology. BPW, in collaboration with SmartSynch, has begun deploying wireless-enabled smart meters, which will operate on a new municipal Wi-Fi network deployed by BPW. Eventually, the utility's 45,000 residential and 6,000 business water and electricity customers in Burbank, California, will be included in the network, and the utility will be able to monitor its entire load using Wi-Fi technology and smart meters.

Because BPW is a water utility as well as an electricity provider, its network will be able to communicate with Wi-Fi-enabled devices in the home that measure water use in real time. With the current drought conditions in the Southeast, and in the West, this can provide an important



The now ever-present wireless antennas dotting the landscape may carry energy-saving messages between your utility and your home appliances.



Yeah, some of those wireless towers are supposed to look like trees.

tool for utilities to reduce water waste in homes. Fred Fletcher, assistant general manager at BPW, says, “We will be able to tell one of our customers as soon as we recognize a problem, ‘Looks like you have a broken pipeline

or a leaky toilet.’ That’s much better than the customer finding out when he or she gets their utility bill at the end of the month.” Utilities, under more and more pressure to supply potable water to consumers with diminishing supplies available from reservoirs and other sources, don’t want to waste water either. And as *Home Energy* readers know, less water use means less energy use moving, heating, and treating the water.

In short, houses are getting smarter; and we will need their smarts to help in the fight against global warming, other environmental degradation, our dependence on fossil fuels, and a water crisis that will only become more critical in the future.

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utility programs

—Wanzhu Du

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For more information:

For more information about SDG&E Smart Meters, go to www.sdge.com/smartmeterv2/index.shtml.

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Find out more about SmartSynch at www.SmartSynch.com.